

Quartz resonator

X26 (2026) — English translation

Error estimation is not required in this task.

Equipment: AC generator Oscilloscope Breadboard Set of jumpers 3 wires BNC-BNC Adapter “BNC-board”
Package with electronic components, which includes: 9 capacitors; quartz resonator, the resonant frequency of which is about 2 MHz ; resistor $R_0 = 5.1\text{ k}\Omega$

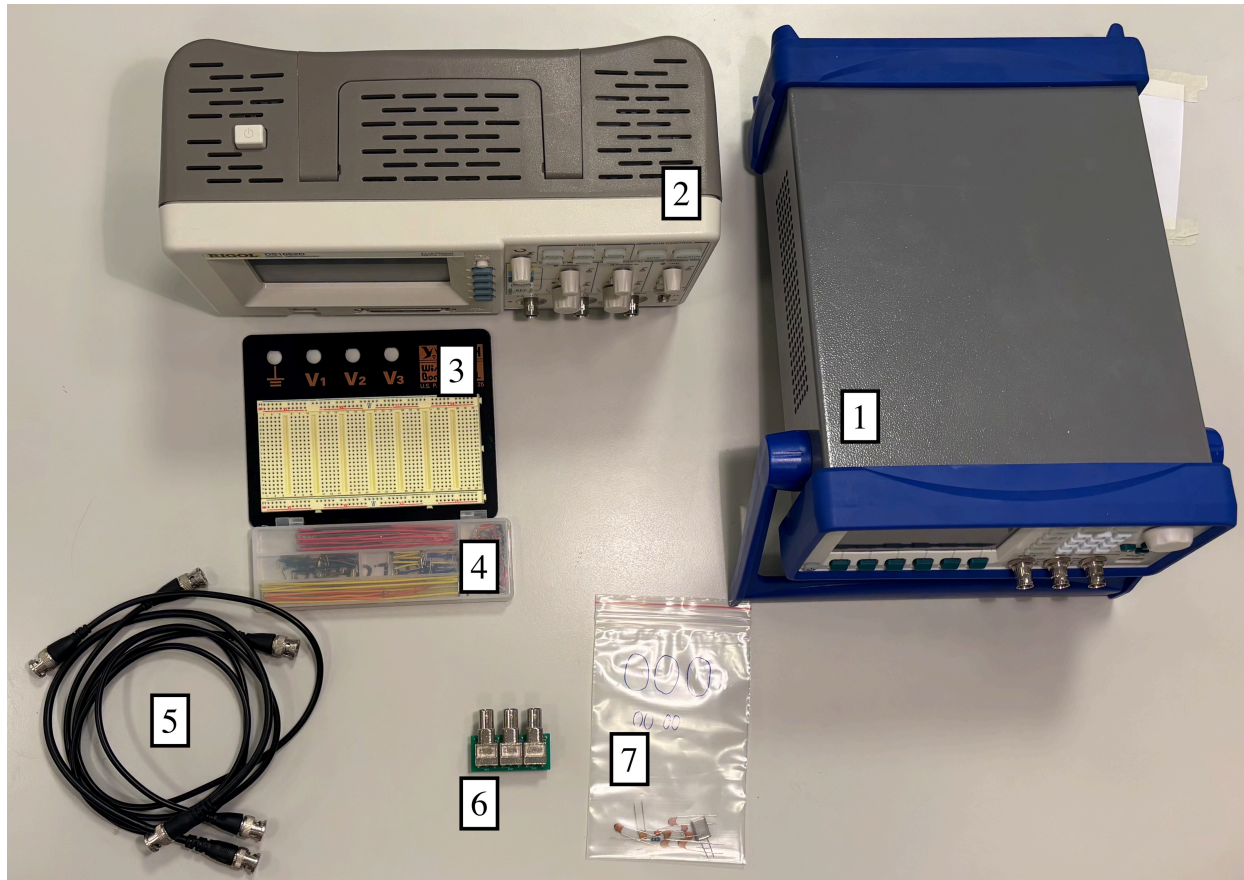


Fig. 1.

Capacitor marking Capacity, pF 6 6.0 10 10 15 15 27 27 33 33 47 47 56 J 56 68 68 82 82

A0 Write down the code for the package containing the electronic components on your answer sheets. Check that it is complete. **0**

A quartz resonator is an electronic component that uses the piezoelectric properties of quartz to generate electrical oscillations with high precision and stability. Based on its behavior in electrical circuits, a quartz resonator can, to a first approximation, be represented in the form of an equivalent electrical circuit shown in the figure. In this problem we will study a quartz resonator as an oscillatory circuit and find its parameters. It is known that the resonant frequency of a quartz resonator is about 2 MHz .

Note. Do not set the double voltage amplitude on the generator $2U_0 > 500\text{ mV}$. Otherwise, the operation of the quartz resonator in the same mode as at lower amplitudes is not guaranteed.

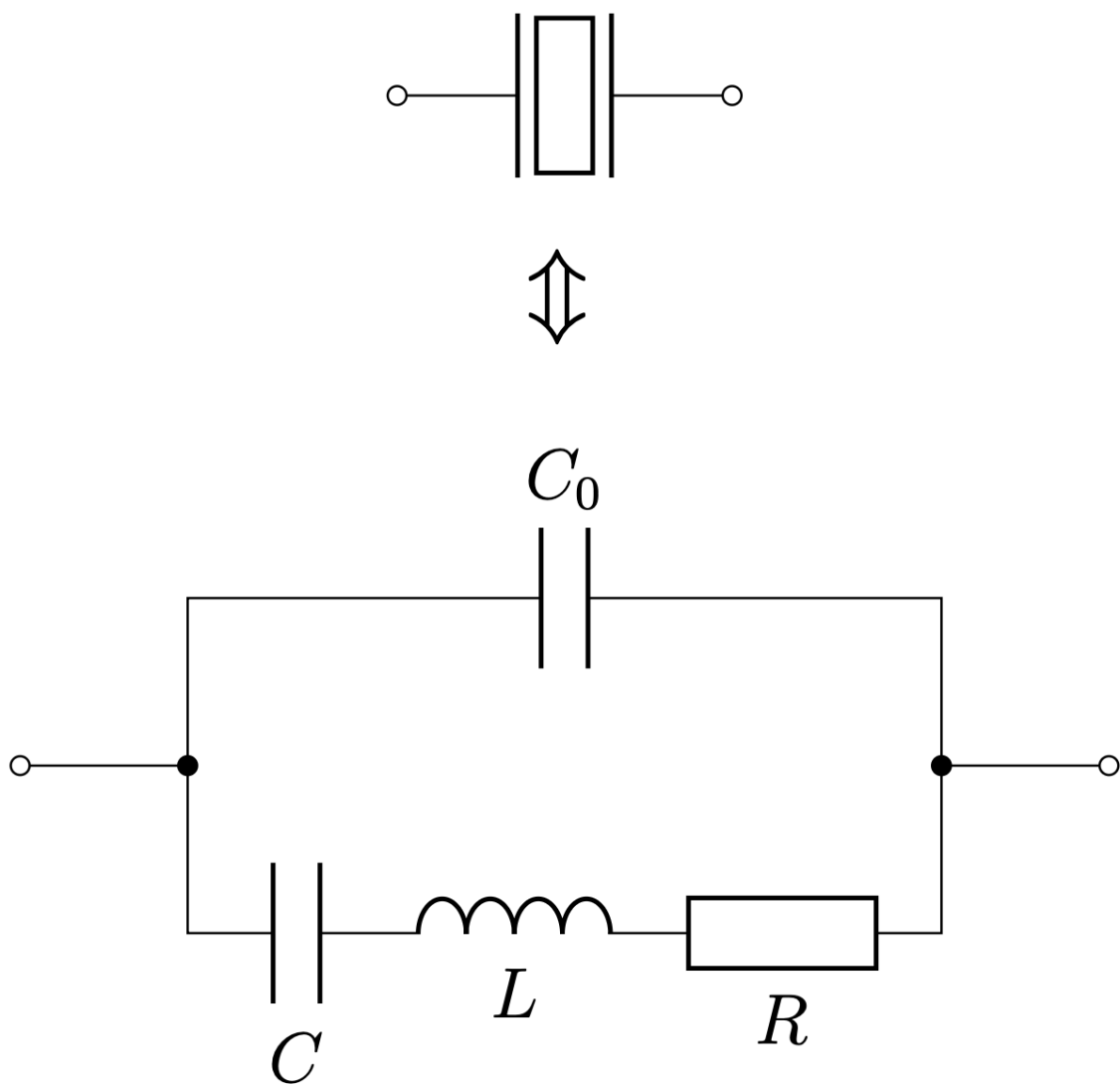


Fig. 2.

For measurements, use the following scheme:

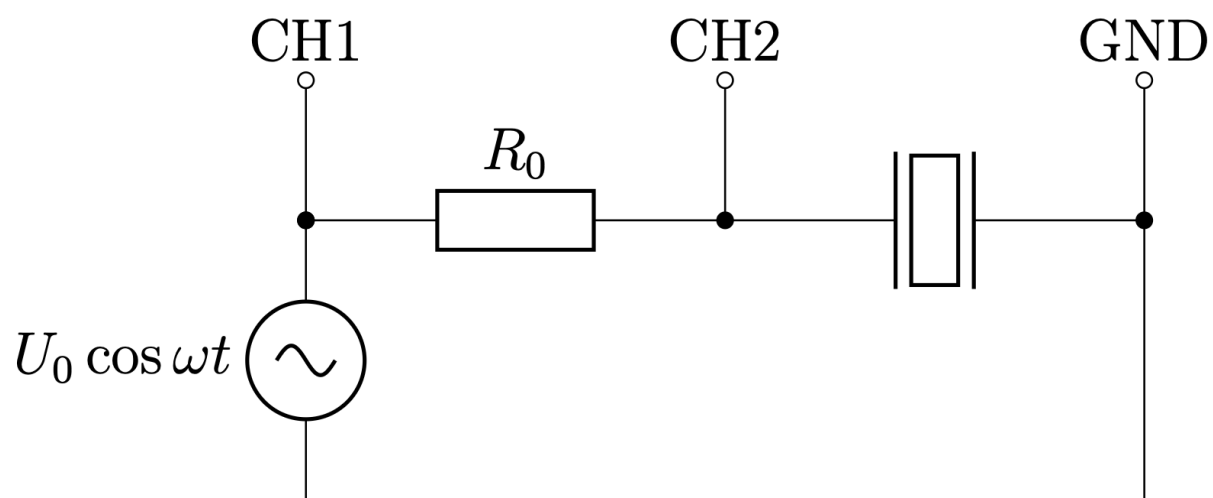


Fig. 3.

A1	Find the signal frequency $f_{\max}$ at which the voltage amplitude across the quartz resonator is maximum. The difference between $f_{\max}$ and $2 M\Gamma y$ is no more than $1 \kappa\Gamma y$.	0.20
A2	Record the ratio of the voltage amplitudes on the two channels of the oscilloscope $\frac{U_{CH2}}{U_{CH1}}$ at the signal frequency $f_{\max}$.	0.05
A3	Remove the dependence of $2U_{CH2}$ and φ on the frequency f in the range $[f_{\max} - 250 \Gamma y; f_{\max} + 250 \Gamma y]$, where φ is the phase difference between the signals from channels 2 and 1. Measure at least 30 points from the given frequency range.	2.80
A4	Plot a graph of the quartz resonator impedance modulus $ Z $ versus frequency f over the measured range.	2.50
A5	From the plotted graph $ Z (f)$, find the frequencies f_{pe3} and f_{anti} at which $ Z $ is maximum and minimum, respectively.	0.10

We will call the frequency f_{pe3} the resonance frequency, and the frequency f_{anti} the antiresonance frequency. The impedance of a quartz resonator can be written as:

$$Z(\omega) = \frac{\omega_s^2 - \omega^2 + i\omega\alpha}{i\omega C_0(\omega_p^2 - \omega^2 + i\omega\alpha)},$$

where ω is the cyclic frequency of the signal; $\omega_s, \omega_p, \alpha$ - do not depend on ω .

A6	Express $\omega_s, \omega_p, \alpha$ in terms of R, L, C, C_0 .	0.30
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Further in this problem we will assume that the following approximations are valid near resonance:

$$Z(\omega) \approx \frac{\omega_s^2 - \omega^2 + i\omega_p\alpha}{i\omega_p C_0(\omega_p^2 - \omega^2 + i\omega_p\alpha)}; \quad \alpha \ll \omega_p - \omega_s.$$

A7	Using the approximations described above, express $ Z ^2(\omega)$ in terms of $\alpha, \omega, \omega_s, \omega_p, C_0$.	0.10
A8	Using the approximations described above, express f_{pe3} and f_{anti} in terms of ω_s, ω_p .	0.20

Further, we can consider it known that within the framework of the described approximations, the following equalities hold:

$$\varphi(f_{anti}) = \varphi(f_{res}) = 0,$$

where $\varphi(f)$ is the dependence of the phase difference between the signals from channels 2 and 1 on frequency. When connecting capacitor C_{ext} parallel to the resonator, the resonance frequency f_{pe3} will change.

A9	Remove the dependence of the resonance frequencies f_{pe3} and antiresonance f_{anti} of the quartz resonator on the external capacitance C_{ext} . As an external capacitance C_{ext} , it is enough to use each of the provided capacitors separately, without combining them.	0.45
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A10	Plot a linearized graph of C_{ext} versus $f_{\text{рез}}$ and $f_{\text{анти}}$.	0.80
A11	From the results of points A6, A10, obtain the values of L, C, C_0 .	1.30
A12	From the results of points A4, A6, A10, obtain the value R .	1.20

Source: <https://pho.rs/p/4788>